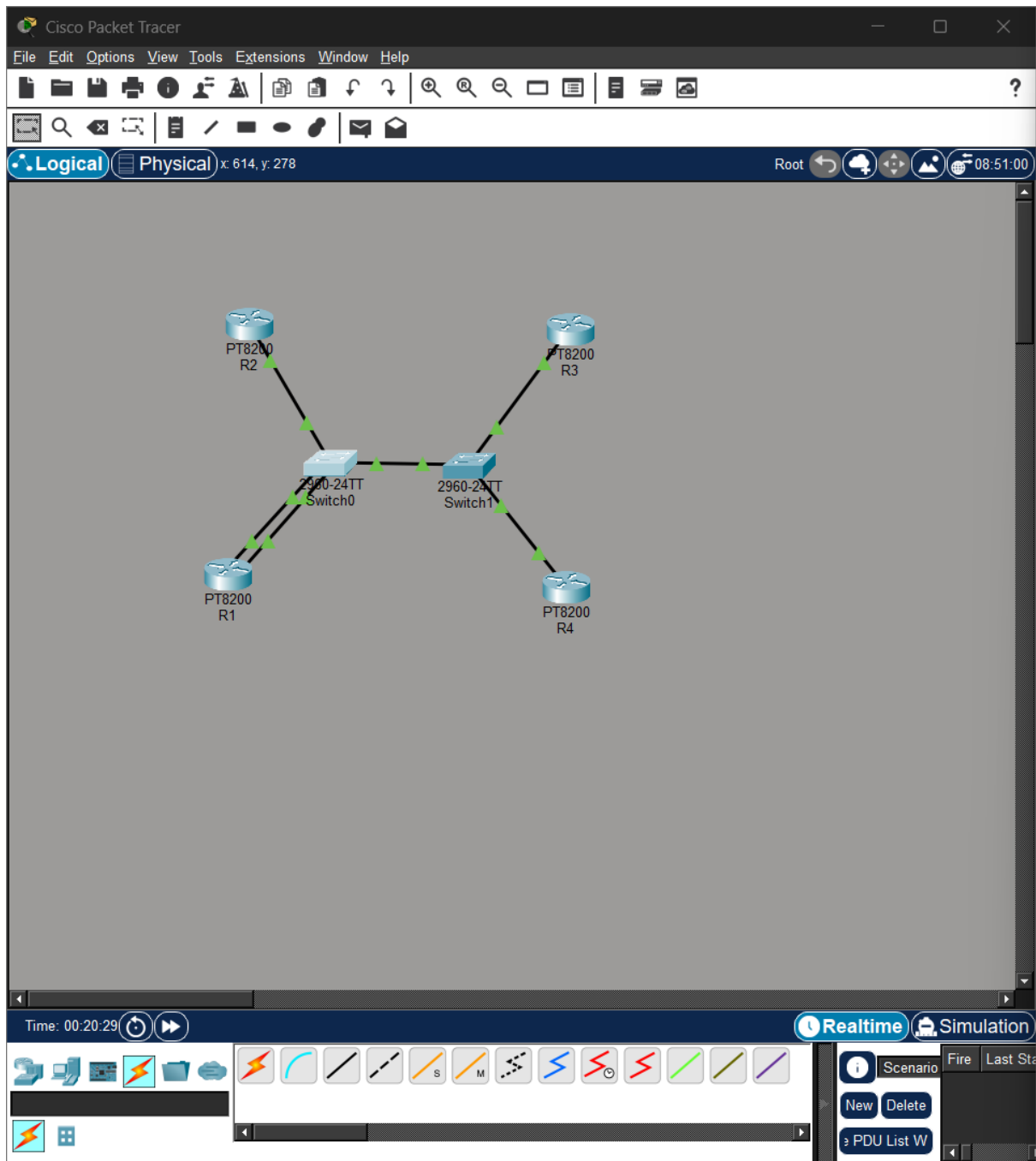
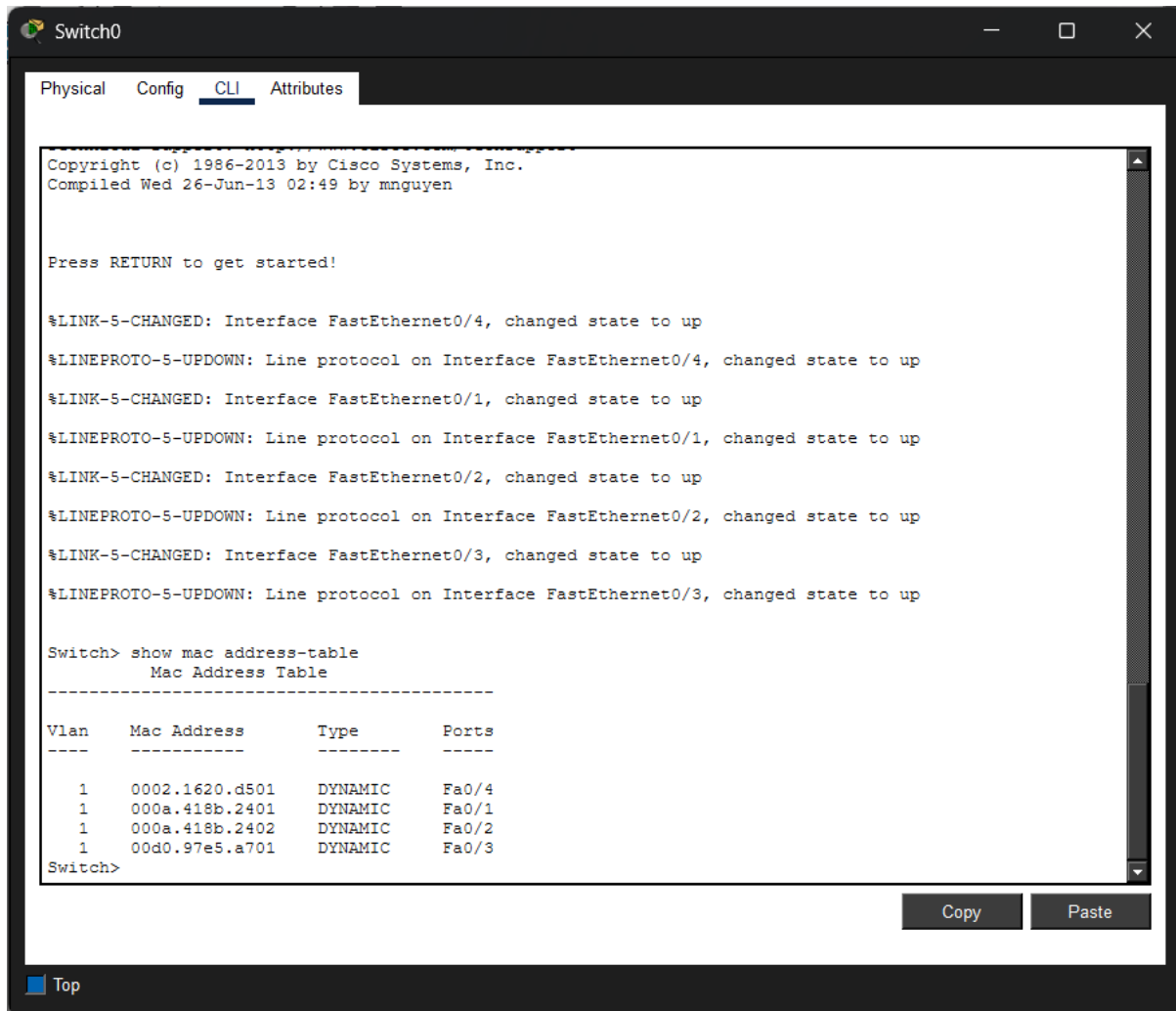




5.



The screenshot shows a Cisco Switch CLI window titled "Switch0". The "CLI" tab is selected. The output shows the following:

```
Copyright (c) 1986-2013 by Cisco Systems, Inc.  
Compiled Wed 26-Jun-13 02:49 by mnnguyen  
  
Press RETURN to get started!  
  
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up  
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up  
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up  
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up  
  
Switch> show mac address-table  
      Mac Address Table  
-----  
Vlan  Mac Address      Type      Ports  
----  -  
1     0002.1620.d501      DYNAMIC   Fa0/4  
1     000a.418b.2401      DYNAMIC   Fa0/1  
1     000a.418b.2402      DYNAMIC   Fa0/2  
1     00d0.97e5.a701      DYNAMIC   Fa0/3  
Switch>
```

At the bottom right, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

6. R1: 000a.418b.2401

```
encryption
Press RETURN to get started!

Router> show mac address-table
      ^
% Invalid input detected at '^' marker.

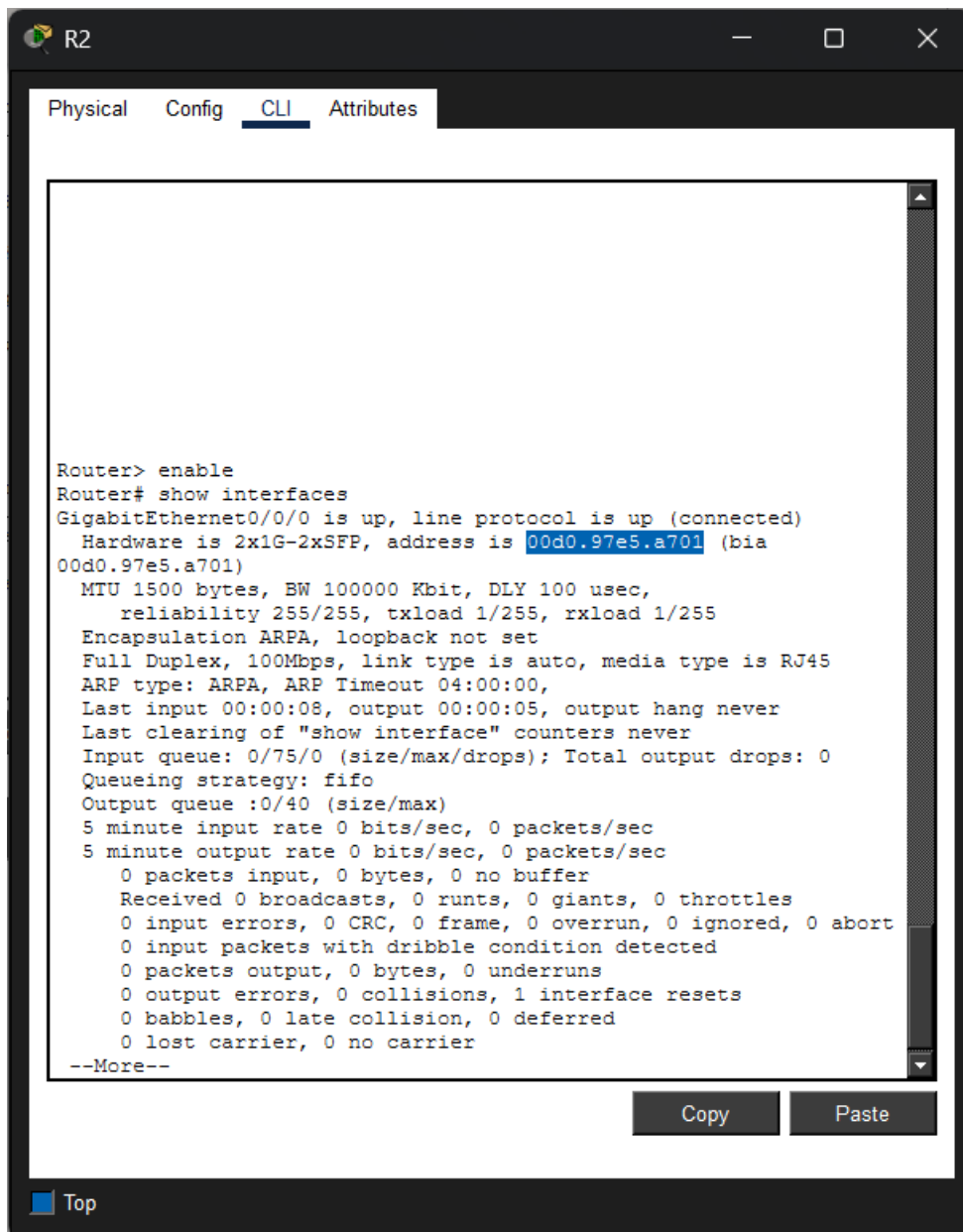
Router> enable
Router# show interfaces
GigabitEthernet0/0/0 is up, line protocol is up (connected)
  Hardware is 2x1G-2xSFP, address is 000a.418b.2401 (bia
000a.418b.2401)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier

Router#
```

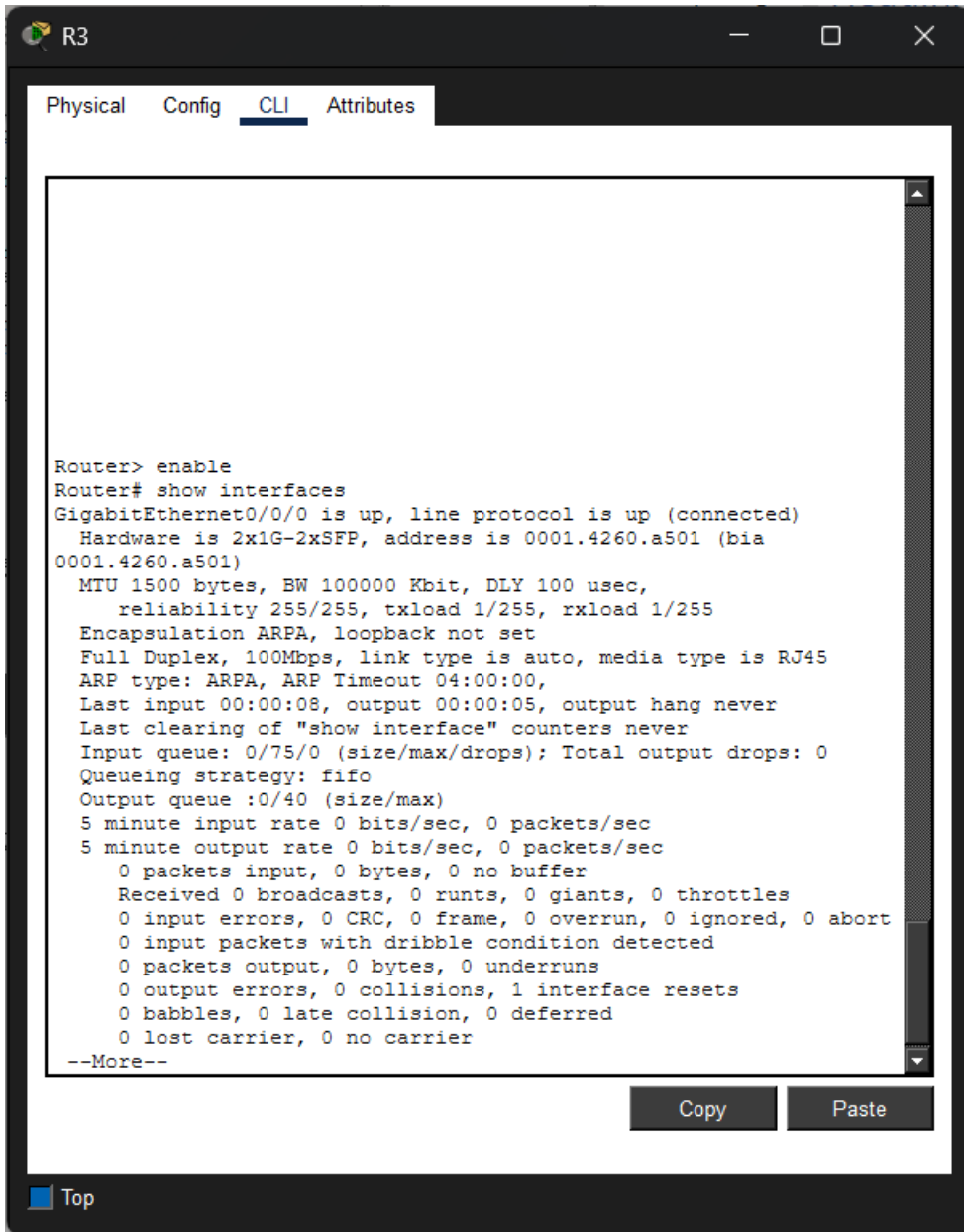
Copy Paste

Top

R2: 00d0.97e5.a701



R3: 0001.4260.a501

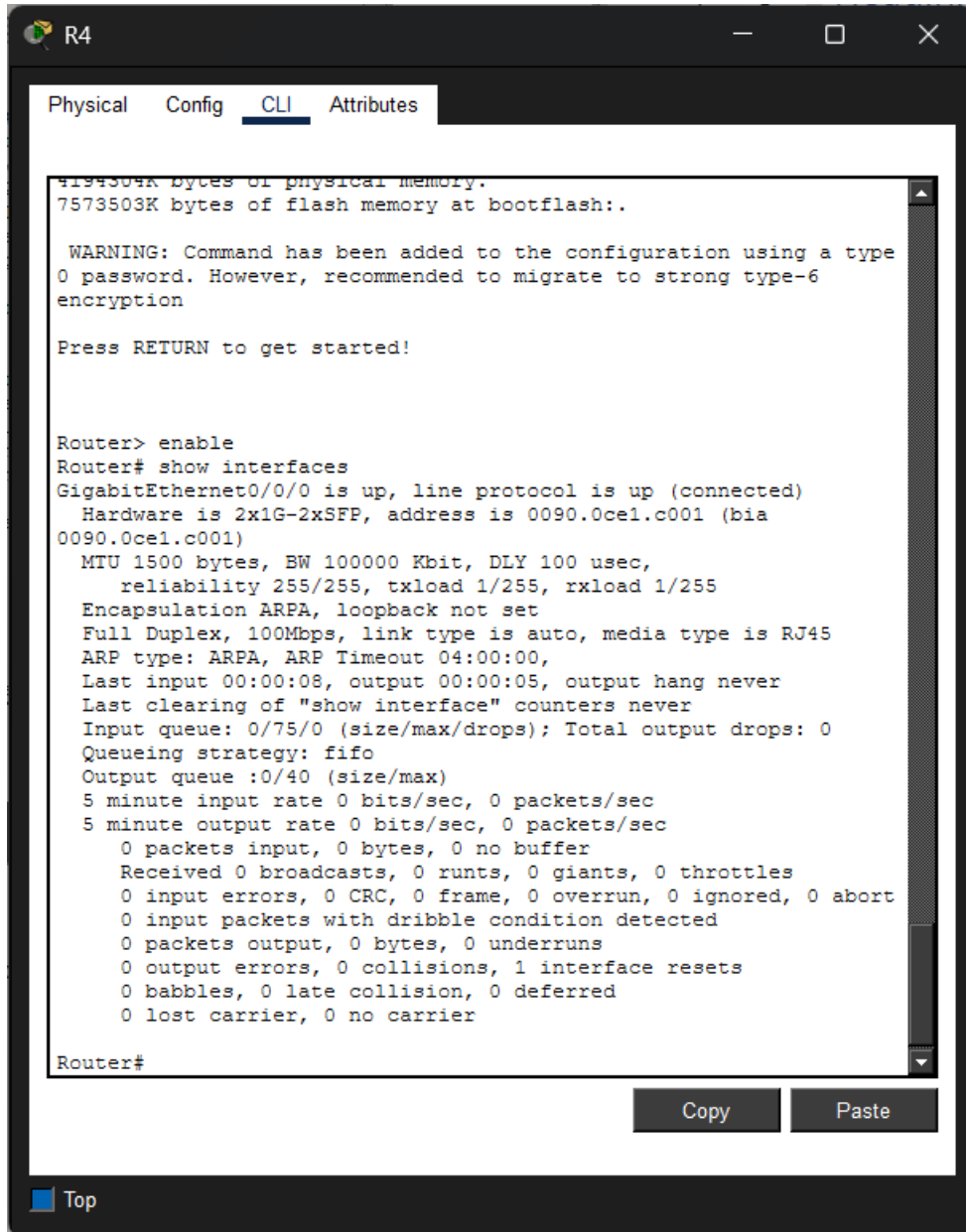


```
Router> enable
Router# show interfaces
GigabitEthernet0/0/0 is up, line protocol is up (connected)
  Hardware is 2x1G-2xSFP, address is 0001.4260.a501 (bia
0001.4260.a501)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
  --More--
```

Copy Paste

Top

R4: 0090.0ce1.c001



7. Ping from R1 to R2

```
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#no ip address
Router(config-if)#no ip address
Router(config-if)#no ip address
Router(config-if)#no ip address
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#no ip address
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 13:32:44.3232: SYS-5-CONFIG_I: Configured from console by
console
Router# ping 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2
seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/1/3
ms

Router#
Router# ping 192.168.12.2

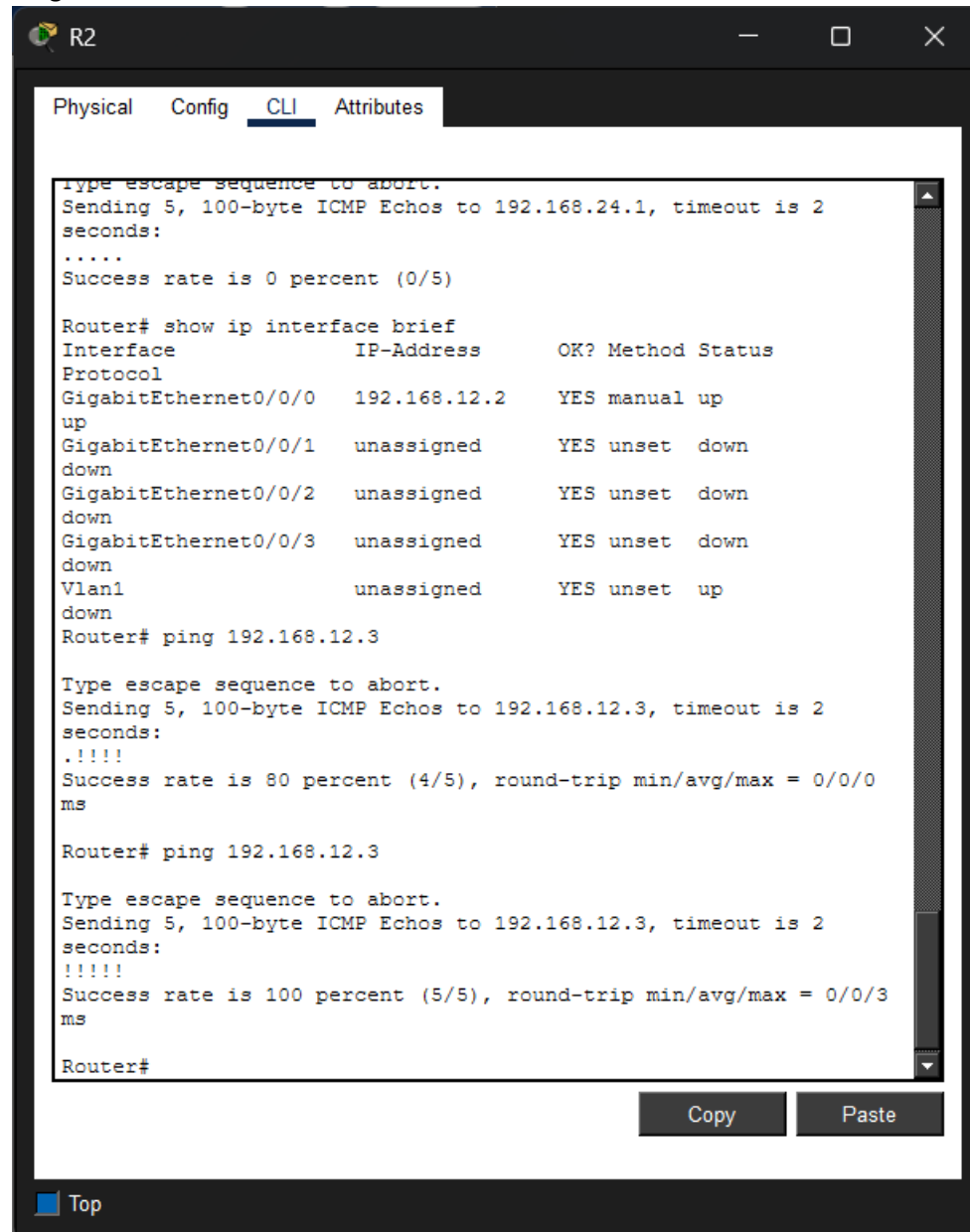
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2
seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0
ms

Router#
```

Copy Paste

Top

Ping from R2 to R3:

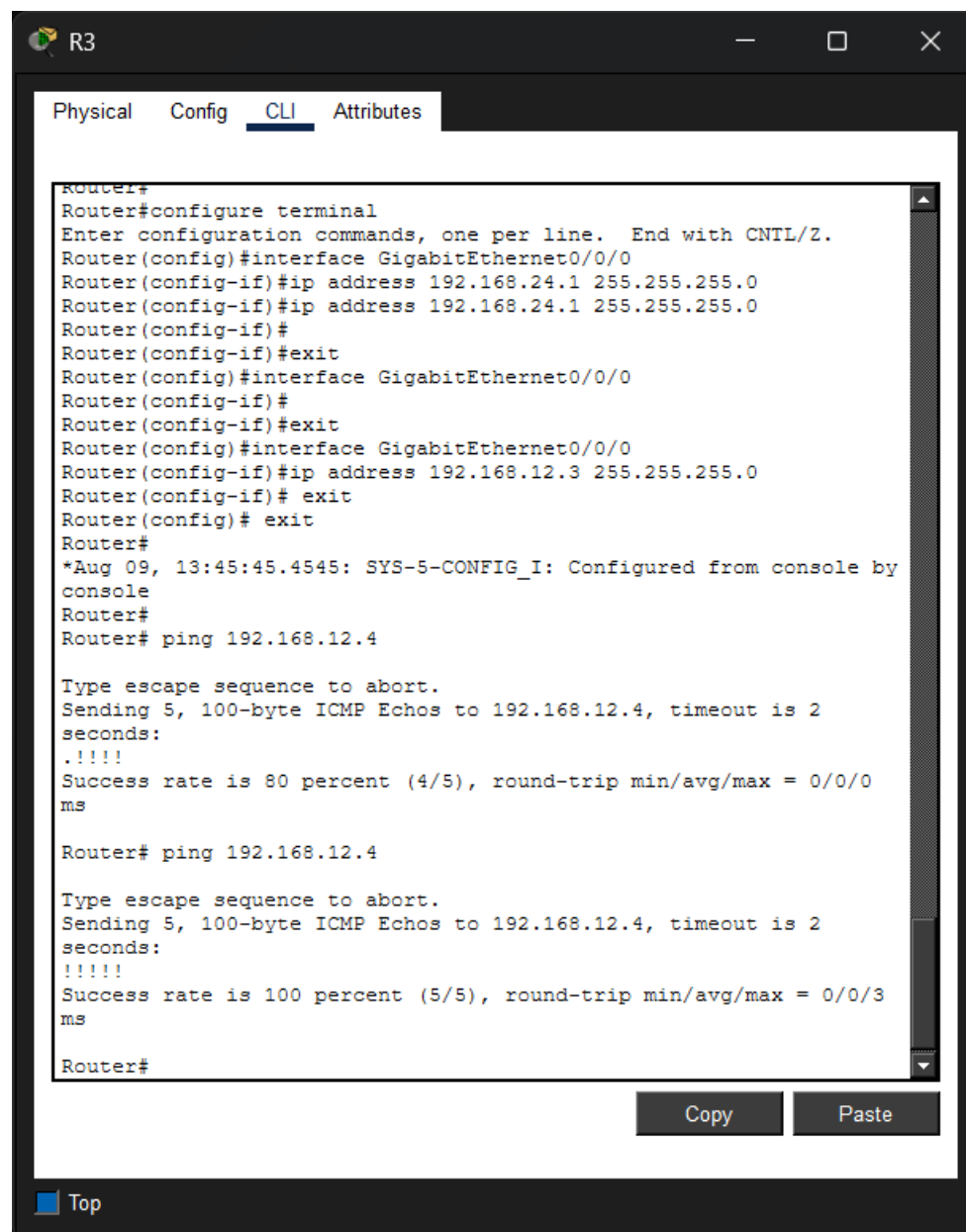


The screenshot shows a Cisco Packet Tracer CLI window for Router R2. The window has tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the following text:

```
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.24.1, timeout is 2  
seconds:  
.....  
Success rate is 0 percent (0/5)  
  
Router# show ip interface brief  
Interface          IP-Address      OK? Method Status  
Protocol  
GigabitEthernet0/0/0 192.168.12.2    YES manual up  
up  
GigabitEthernet0/0/1 unassigned      YES unset  down  
down  
GigabitEthernet0/0/2 unassigned      YES unset  down  
down  
GigabitEthernet0/0/3 unassigned      YES unset  down  
down  
Vlan1                unassigned      YES unset  up  
down  
Router# ping 192.168.12.3  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.12.3, timeout is 2  
seconds:  
.!!!!  
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0  
ms  
  
Router# ping 192.168.12.3  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.12.3, timeout is 2  
seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/3  
ms  
  
Router#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' button.

Ping from R3 to R4:



The screenshot shows a Cisco Router CLI window titled 'R3'. The window has tabs for 'Physical', 'Config', 'CLI', and 'Attributes', with 'CLI' selected. The CLI shows the following commands and output:

```
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.24.1 255.255.255.0
Router(config-if)#ip address 192.168.24.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.3 255.255.255.0
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 13:45:45.4545: SYS-5-CONFIG_I: Configured from console by
console
Router#
Router# ping 192.168.12.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.4, timeout is 2
seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0
ms

Router# ping 192.168.12.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.4, timeout is 2
seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/3
ms

Router#
```

At the bottom of the window, there are 'Copy' and 'Paste' buttons, and a 'Top' button with a blue square icon.

8.

SW1

Physical Config **CLI** Attributes

```
Switch con0 is now available

Press RETURN to get started.

Switch> show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
-----
1       0002.1620.d501   DYNAMIC Fa0/4
1       000a.418b.2401   DYNAMIC Fa0/1
1       000a.418b.2402   DYNAMIC Fa0/2
1       00d0.97e8.a701   DYNAMIC Fa0/3
Switch>
```

Copy Paste

Top

SW2

Physical Config **CLI** Attributes

```
Cisco IOS Software, C2960 Software (C2960-K9M), Version
15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up

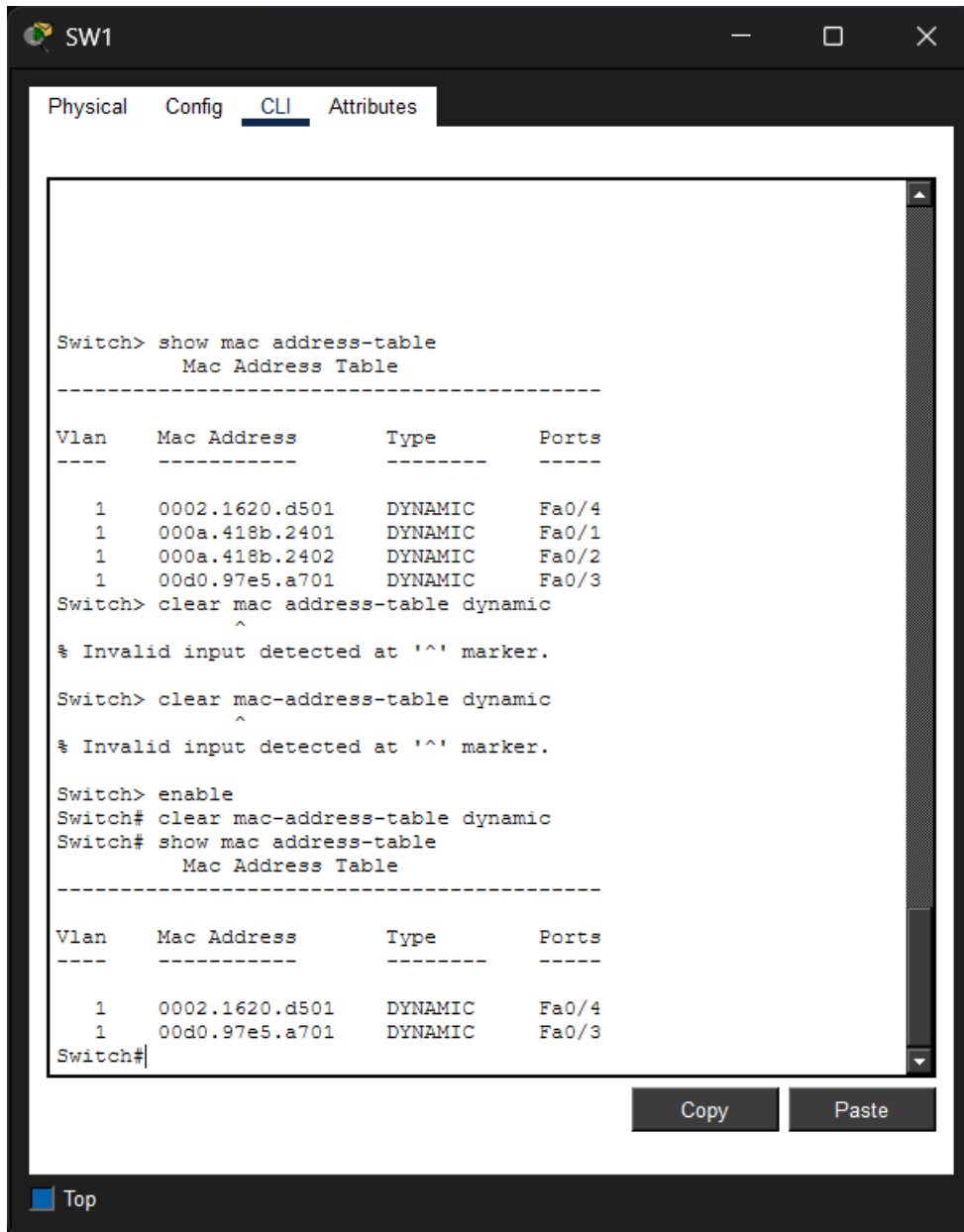
Switch> show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
-----
1       0001.4260.a501   DYNAMIC Fa0/2
1       0050.0f5e.ad04   DYNAMIC Fa0/1
1       0090.0cel.c001   DYNAMIC Fa0/3
Switch>
```

Copy Paste

Top

9. Clear Mac address and check again:

SW1:



The screenshot shows a Cisco switch CLI window titled 'SW1'. The 'CLI' tab is selected. The user enters the command 'show mac address-table', which displays a table of four dynamic MAC addresses. Then, the user enters 'clear mac address-table dynamic', which results in an error message: '% Invalid input detected at '^' marker.' The user then enters 'clear mac-address-table dynamic', which also results in the same error message. Finally, the user enters 'enable' to enter privileged EXEC mode, followed by 'clear mac-address-table dynamic'. This command is successful, and the user then enters 'show mac address-table' again, which now displays a table with only two dynamic MAC addresses. The window includes a 'Copy' button and a 'Paste' button at the bottom right, and a 'Top' link at the bottom left.

```
SW1> show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.1620.d501   DYNAMIC   Fa0/4
1       000a.418b.2401   DYNAMIC   Fa0/1
1       000a.418b.2402   DYNAMIC   Fa0/2
1       00d0.97e5.a701   DYNAMIC   Fa0/3
Switch> clear mac address-table dynamic
      ^
% Invalid input detected at '^' marker.

Switch> clear mac-address-table dynamic
      ^
% Invalid input detected at '^' marker.

Switch> enable
Switch# clear mac-address-table dynamic
Switch# show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.1620.d501   DYNAMIC   Fa0/4
1       00d0.97e5.a701   DYNAMIC   Fa0/3
Switch#
```

SW2:

SW2

Physical
Config
CLI
Attributes

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up

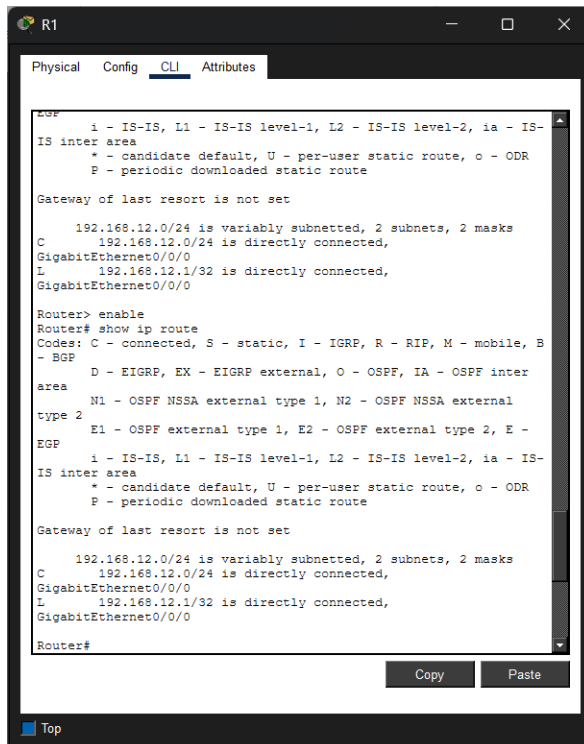
Switch> show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.4260.a501    DYNAMIC   Fa0/2
1       0050.0f5e.ad04    DYNAMIC   Fa0/1
1       0090.0ce1.c001    DYNAMIC   Fa0/3
Switch> enable
Switch# clear mac address-table dynamic
Switch# show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.4260.a501    DYNAMIC   Fa0/2
1       0050.0f5e.ad04    DYNAMIC   Fa0/1
Switch#

```

Copy
Paste

Top

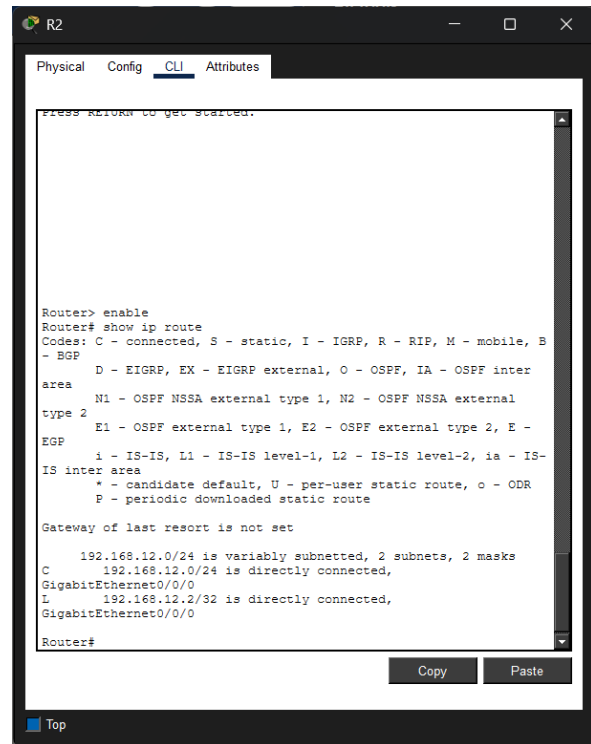
10.



```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
       type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
       EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
       IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

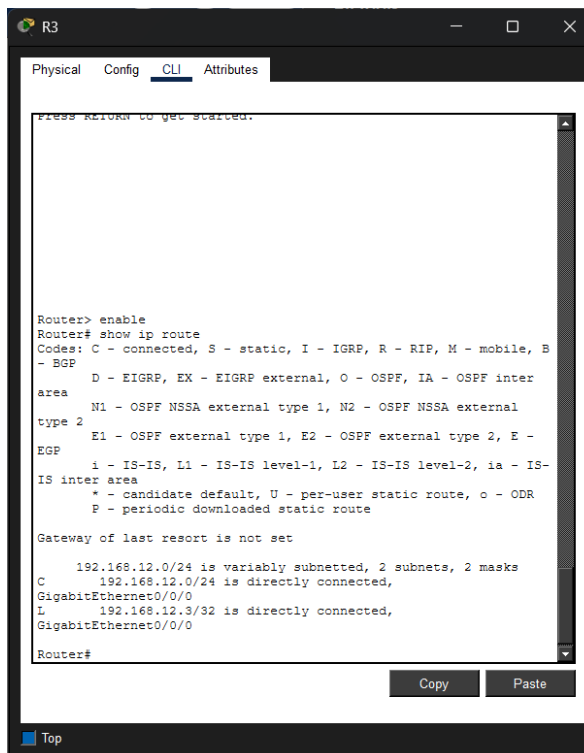
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.1/32 is directly connected,
GigabitEthernet0/0/0
Router#
```



```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
       type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
       EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
       IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

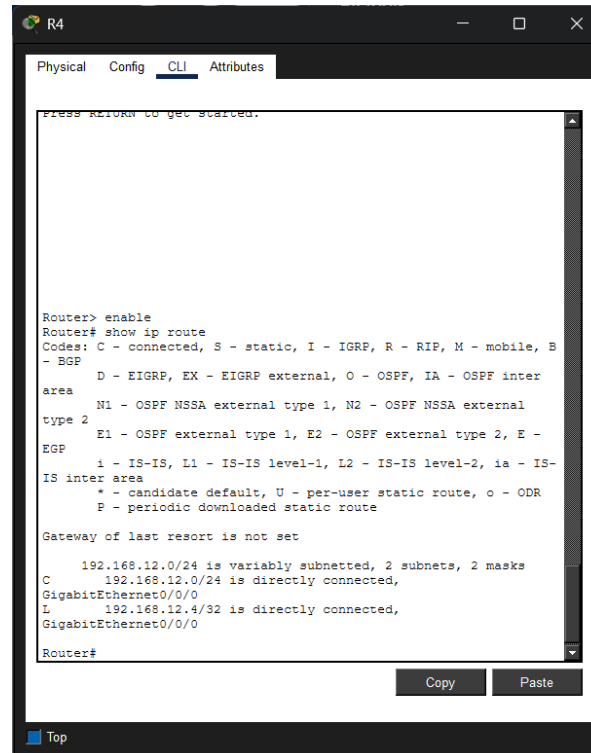
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.2/32 is directly connected,
GigabitEthernet0/0/0
Router#
```



```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
       type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
       EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
       IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.3/32 is directly connected,
GigabitEthernet0/0/0
Router#
```



```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
       type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
       EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
       IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.4/32 is directly connected,
GigabitEthernet0/0/0
Router#
```

11. Assigning IP addresses:

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:19:10.1919: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0/0	192.168.12.1	YES	manual	up
GigabitEthernet0/0/1	unassigned	YES	unset	up
GigabitEthernet0/0/2	unassigned	YES	unset	down
GigabitEthernet0/0/3	unassigned	YES	unset	down
Vlan1	unassigned	YES	unset	up

```

IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L 192.168.12.2/32 is directly connected,
GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:20:36.2020: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0/0	192.168.12.2	YES	manual	up
GigabitEthernet0/0/1	unassigned	YES	unset	down
GigabitEthernet0/0/2	unassigned	YES	unset	down
GigabitEthernet0/0/3	unassigned	YES	unset	down
Vlan1	unassigned	YES	unset	up

```

IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L 192.168.12.3/32 is directly connected,
GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.3 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:22:38.2222: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0/0	192.168.12.3	YES	manual	up
GigabitEthernet0/0/1	unassigned	YES	unset	down
GigabitEthernet0/0/2	unassigned	YES	unset	down
GigabitEthernet0/0/3	unassigned	YES	unset	down
Vlan1	unassigned	YES	unset	up

```

IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

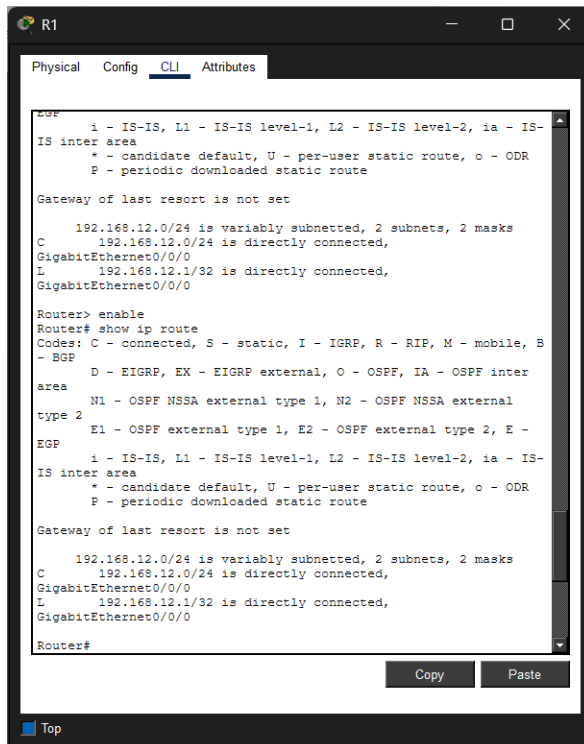
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L 192.168.12.4/32 is directly connected,
GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.4 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:23:32.2323: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0/0	192.168.12.4	YES	manual	up
GigabitEthernet0/0/1	unassigned	YES	unset	down
GigabitEthernet0/0/2	unassigned	YES	unset	down
GigabitEthernet0/0/3	unassigned	YES	unset	down
Vlan1	unassigned	YES	unset	up

12.

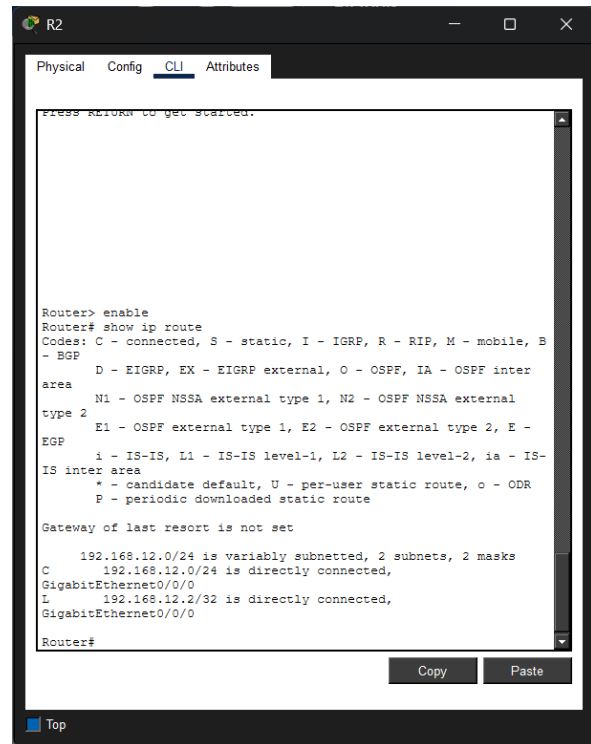


```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.1/32 is directly connected,
GigabitEthernet0/0/0
Router#
```

Copy Paste

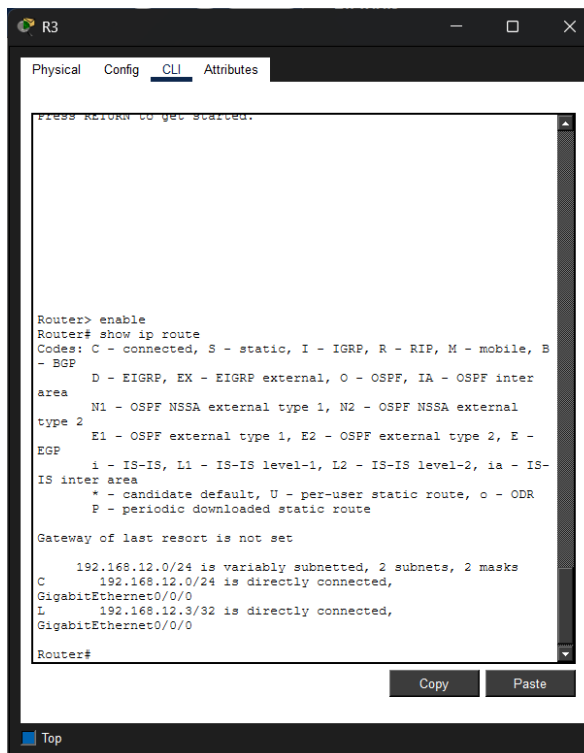


```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.2/32 is directly connected,
GigabitEthernet0/0/0
Router#
```

Copy Paste

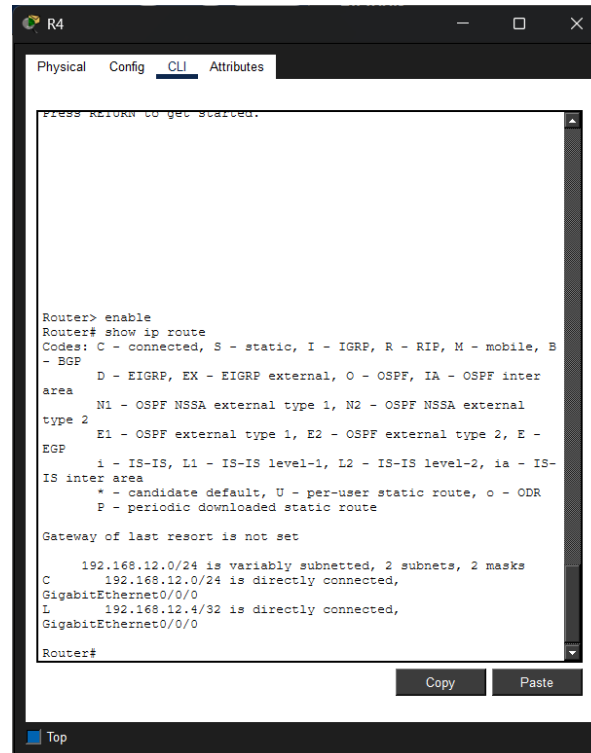


```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.3/32 is directly connected,
GigabitEthernet0/0/0
Router#
```

Copy Paste



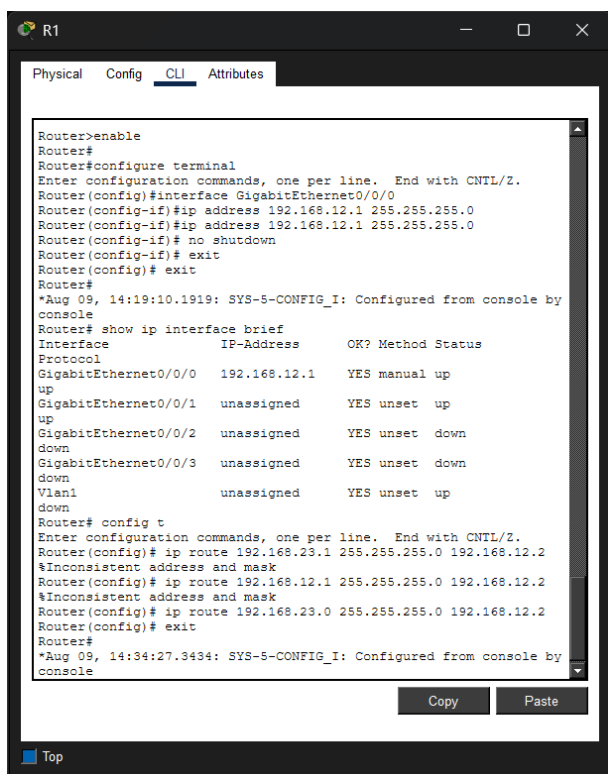
```
Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.4/32 is directly connected,
GigabitEthernet0/0/0
Router#
```

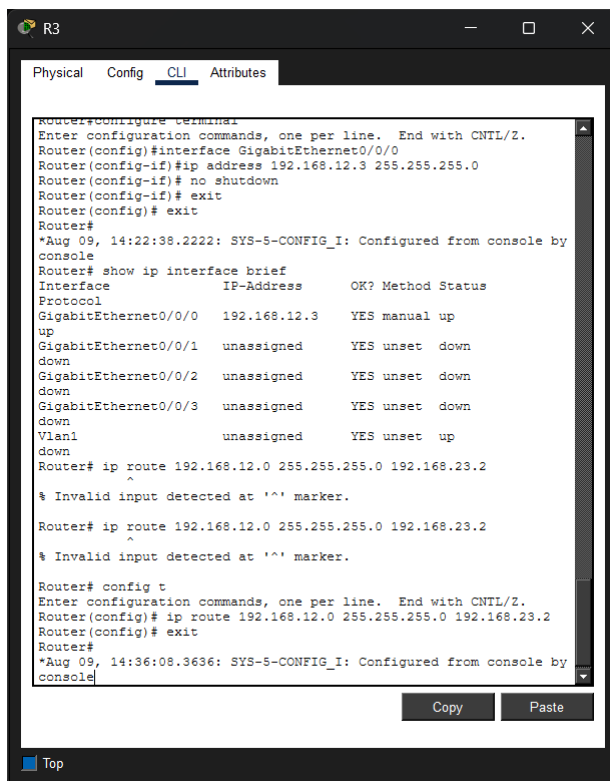
Copy Paste

13.



The screenshot shows the CLI interface for Router R1. The 'CLI' tab is selected. The user has entered the following commands: `enable`, `configure terminal`, `interface GigabitEthernet0/0/0`, `ip address 192.168.12.1 255.255.255.0`, `no shutdown`, and `exit`. The router has confirmed the configuration. The user then enters `show ip interface brief`, which displays a table of interface status. The table shows that GigabitEthernet0/0/0 is up and has the IP address 192.168.12.1. The other interfaces (0/0/1, 0/0/2, 0/0/3, and Vlan1) are down and unassigned. The user then enters `config t` and `ip route 192.168.23.1 255.255.255.0 192.168.12.2`, but receives an error message: `%Inconsistent address and mask`. The user then enters `ip route 192.168.12.1 255.255.255.0 192.168.12.2`, which also results in an error. Finally, the user enters `ip route 192.168.23.0 255.255.255.0 192.168.12.2`, which is accepted. The user then enters `exit` and `exit` to return to the privileged EXEC mode. The router has confirmed the configuration. The user then enters `show ip interface brief` again, which displays the same table as before. The user then enters `config t` and `ip route 192.168.12.0 255.255.255.0 192.168.23.2`, but receives an error message: `% Invalid input detected at '^' marker.` The user then enters `ip route 192.168.12.0 255.255.255.0 192.168.23.2`, which also results in an error. Finally, the user enters `config t` and `ip route 192.168.12.0 255.255.255.0 192.168.23.2`, which is accepted. The user then enters `exit` and `exit` to return to the privileged EXEC mode. The router has confirmed the configuration. The user then enters `show ip interface brief` again, which displays the same table as before.

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
*Aug 09, 14:19:10.1919: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0 192.168.12.1   YES manual up
up
GigabitEthernet0/0/1 unassigned      YES unset  up
up
GigabitEthernet0/0/2 unassigned      YES unset  down
down
GigabitEthernet0/0/3 unassigned      YES unset  down
down
Vlan1              unassigned      YES unset  up
down
Router# config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# ip route 192.168.23.1 255.255.255.0 192.168.12.2
%Inconsistent address and mask
Router(config)# ip route 192.168.12.1 255.255.255.0 192.168.12.2
%Inconsistent address and mask
Router(config)# ip route 192.168.23.0 255.255.255.0 192.168.12.2
Router(config)# exit
Router#
*Aug 09, 14:34:27.3434: SYS-5-CONFIG_I: Configured from console by
console
```



The screenshot shows the CLI interface for Router R3. The 'CLI' tab is selected. The user has entered the following commands: `configure terminal`, `interface GigabitEthernet0/0/0`, `ip address 192.168.12.3 255.255.255.0`, `no shutdown`, and `exit`. The router has confirmed the configuration. The user then enters `show ip interface brief`, which displays a table of interface status. The table shows that GigabitEthernet0/0/0 is up and has the IP address 192.168.12.3. The other interfaces (0/0/1, 0/0/2, 0/0/3, and Vlan1) are down and unassigned. The user then enters `ip route 192.168.12.0 255.255.255.0 192.168.23.2`, but receives an error message: `% Invalid input detected at '^' marker.` The user then enters `ip route 192.168.12.0 255.255.255.0 192.168.23.2`, which also results in an error. Finally, the user enters `config t` and `ip route 192.168.12.0 255.255.255.0 192.168.23.2`, which is accepted. The user then enters `exit` and `exit` to return to the privileged EXEC mode. The router has confirmed the configuration. The user then enters `show ip interface brief` again, which displays the same table as before.

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.3 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
*Aug 09, 14:22:38.2222: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0 192.168.12.3   YES manual up
up
GigabitEthernet0/0/1 unassigned      YES unset  down
down
GigabitEthernet0/0/2 unassigned      YES unset  down
down
GigabitEthernet0/0/3 unassigned      YES unset  down
down
Vlan1              unassigned      YES unset  up
down
Router# ip route 192.168.12.0 255.255.255.0 192.168.23.2
^
% Invalid input detected at '^' marker.
Router# ip route 192.168.12.0 255.255.255.0 192.168.23.2
^
% Invalid input detected at '^' marker.
Router# config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# ip route 192.168.12.0 255.255.255.0 192.168.23.2
Router(config)# exit
Router#
*Aug 09, 14:36:08.3636: SYS-5-CONFIG_I: Configured from console by
console
```


14.

```

R1
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B
- BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L    192.168.12.1/32 is directly connected,
GigabitEthernet0/0/0
S    192.168.23.0/24 [1/0] via 192.168.12.2
Router#

```

```

R2
Press RETURN to get started.

Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B
- BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L    192.168.12.2/32 is directly connected,
GigabitEthernet0/0/0
Router#

```

```

R3
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B
- BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L    192.168.12.3/32 is directly connected,
GigabitEthernet0/0/0
Router#

```

```

R4
Press RETURN to get started.

Router> enable
Router# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B
- BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L    192.168.12.4/32 is directly connected,
GigabitEthernet0/0/0
Router#

```

15. Yes it does because it's configured correctly.